

Compatibilization of Polyethylene and Polypropylene via Dynamic Cross-Linking

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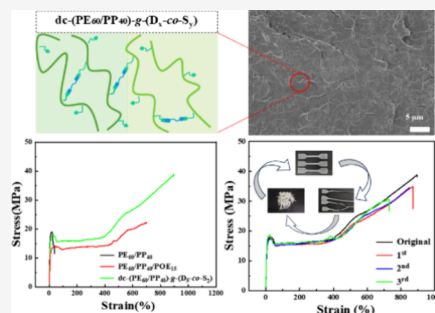


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ABSTRACT: Despite similar hydrocarbon compositions, polyethylene (PE) and polypropylene (PP) are mutually immiscible, posing challenges to their recycling. Consequently, significant research efforts have been made on PE/PP compatibilization. In this study, we transformed physical blends of PE and PP into dynamic covalent polymer networks by taking advantage of reversible boronate ester bonds. The dynamically cross-linked network “welds” PE and PP phases together, enabling PE–PP interfacial compatibilization through morphological control and converting the otherwise brittle PE and PP blend into a tough PE/PP hybrid plastic.



1. INTRODUCTION

Polyolefins, typically polyethylene (PE) and polypropylene (PP), account for more than half of global plastic production due to their excellent physical properties and high cost-effectiveness. In 2021, global plastic production reached 390 million tons,¹ and its consumption continues to increase year by year. Since polyolefins are primarily used as disposable packaging materials, this means that millions of tons of polyolefin waste are generated annually.^{2–4} Therefore, the effective collection and recycling of polyolefins are of critical importance.⁵ Large-scale recycling of polyolefins is primarily carried out mechanically; however, the potential incompatibility between these polymers makes direct blending insufficient for effective recycling and reuse, inevitably reducing product quality.^{4,6–10} Thus, developing compatibilizers and efficient compatibilization systems received much attention in recent years.^{11–14}

Although various methods, both reactive^{15–18} and non-reactive,^{19–23} have been reported for compatibilizing polyolefin blends, the development of a simple and efficient approach for PE/PP blends remains an urgent need. However, current understanding of the compatibilization mechanisms in such blends is still largely limited to molecular chain entanglement, which proves inadequate for significantly improving tensile strength. Recognizing the widespread application of dynamic cross-linking networks in polyolefin modification,^{24–27} we propose introducing dynamic boron–oxygen cross-linking networks at polyolefin blend interfaces to achieve molecular chain rearrangement and improved compatibility.

In this study, we constructed a dynamic covalent network of PE and PP polymers by utilizing PE_{60}/PP_{40} , a blend of PE and PP at 60/40 weight ratio, as base, a boronic ester-function-alized methyl methacrylate (DBMA) as functional monomer,

styrene (St) as comonomer, dicumyl peroxide (DCP) as initiator, and bis-dioxaborolane as cross-linker via metathesis reaction, resulting in a dynamically cross-linked, tough PE_{60}/PP_{40} hybrid plastic, $dc-(PE_{60}/PP_{40})-g-(D_x-co-S_y)$, D and S denote DBMA and St, respectively. By tuning the content of DBMA and cross-linker, we achieved strength-ductility balanced $dc-(PE_{60}/PP_{40})-g-(D_5-co-S_2)$ of significantly enhanced tensile strength and elongation at break, as high as 102.57% and 1422.22% respective improvement, with only a slight reduction in Young's modulus (5.95%). The effects of the dynamic boron–oxygen cross-linking network structure on the phase morphology, crystallization behavior and mechanical properties of PE/PP blend were systematically investigated and the relationship between phase morphology and mechanical performance was explored. Furthermore, the incorporation of dynamic covalent networks endowed the resulting PE/PP hybrid plastic with excellent reprocessing and recycling capabilities, extending the life cycle of the material. Our research provides an effective pathway to industrial recycling of PE and PP blends while preserving existing optimized synthesis and processing conditions, thereby preparing a dynamically cross-linked PE_{60}/PP_{40} blend.

2. RESULTS AND DISCUSSION

The dynamic, reconfigurable covalent network of $dc-(PE_{60}/PP_{40})-g-(D_x-co-S_y)$ was constructed by grafting D_x-co-S_y ,

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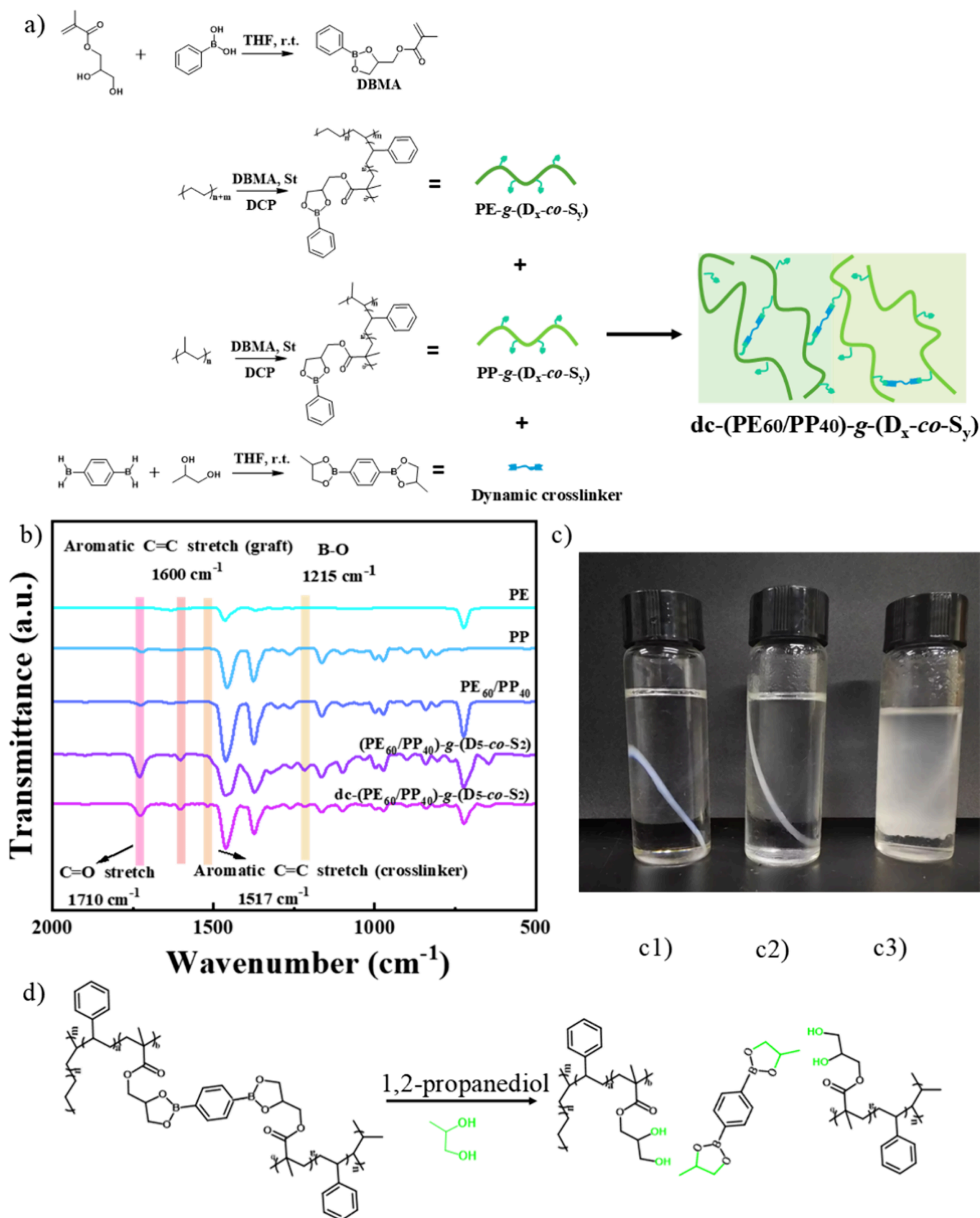


Figure 1. (a) Synthesis of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂). (b) FTIR spectra of PE, PP, (PE₆₀/PP₄₀), (PE₆₀/PP₄₀)-g-(D₅-co-S₂), and dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂). (c1, c2) Optical microscopy image of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) in xylene at 25 and 120 °C, respectively. (c3) Optical microscopy image of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) in a solvent mixture of xylene and 1,2-propanediol at 120 °C. (d) De-cross-linking reaction.

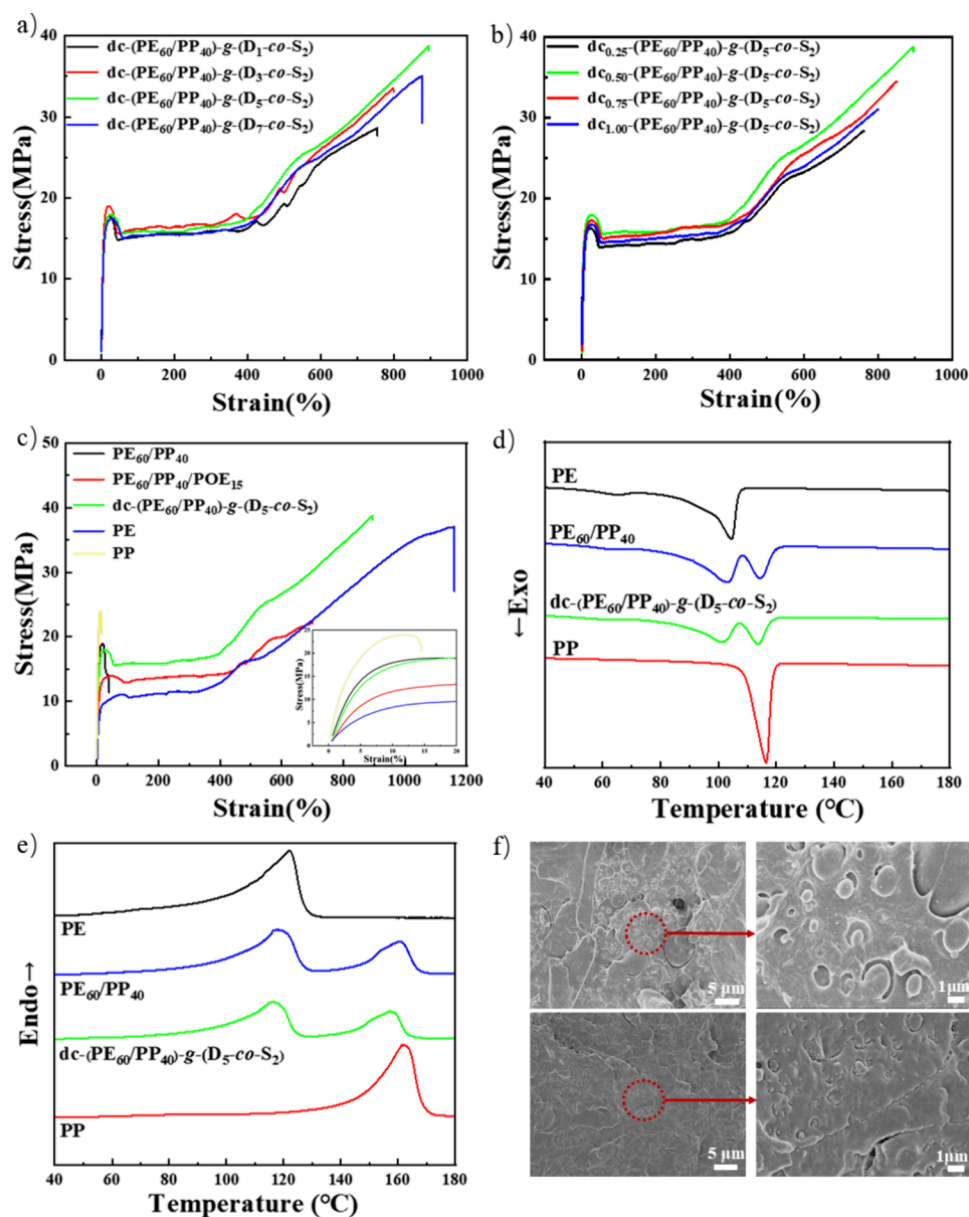


Figure 2. (a) Stress–strain curves of $dc_{0.5}-(PE_{60}/PP_{40})-g-(D_5-co-S_2)$ with varying DBMA contents. (b) Stress–strain curves of $dc_{x}-(PE_{60}/PP_{40})-g-(D_5-co-S_2)$ with varying cross-linker contents. (c) Stress–strain curves of PE, PP, PE_{60}/PP_{40} , $PE_{60}/PP_{40}/POE_{15}$, and $dc-(PE_{60}/PP_{40})-g-(D_5-co-S_2)$. (d) DSC curves of PE, PP, PE_{60}/PP_{40} , and $dc-(PE_{60}/PP_{40})-g-(D_5-co-S_2)$ indicating crystallization temperatures. (e) DSC (second heating cycle) curve of PE, PP, PE_{60}/PP_{40} , and $dc-(PE_{60}/PP_{40})-g-(D_5-co-S_2)$ indicating melting points. (f) SEM images of PE_{60}/PP_{40} (first row) and $dc-(PE_{60}/PP_{40})-g-(D_5-co-S_2)$ (second row).

copolymerized from the complex monomer DBMA/St, onto the polymer backbone in the PE_{60}/PP_{40} blend, followed by reacting with a dioxane-based dynamic cross-linker, 1,4-phenylenediboronic acid ester, via a borate ester complexation reaction (Figure 1a). Using styrene as a comonomer increases the grafting rate of the functional monomer DBMA with the PE_{60}/PP_{40} blend, a well-known technique employed for grafting polar modifiers,²⁸ due primarily to the higher reactivity of styrene, which helps form a stable DBMA/St structure as the target complex monomer for grafting.²⁹

The dioxaborolane maleimide graft exhibited a C=O stretching peak at 1710 cm^{-1} and aromatic C=C stretching peak at 1600 cm^{-1} . Bis-dioxaborolane (i.e., 1,4-phenylenediboronic acid ester) exhibited an aromatic C=C stretching signal at 1517 cm^{-1} . The appearance of the absorption peak at

1215 cm^{-1} can be attributed to the formation of B–O bonds in the PE_{60}/PP_{40} blend (Figure 1b). These FTIR results indicate that DBMA has been successfully grafted onto both PE and PP polymers, which were then cross-linked by 1,4-phenylenediboronic acid ester. $Dc-(PE_{60}/PP_{40})-g-(D_5-co-S_2)$, a representative of such vitrimers, is only partially dissolved in xylene at room temperature and 120 °C (Figure 1c1,c2). The gel content at 25 and 120 °C was 96.95% and 94.46% (Figure S1), respectively, mostly likely because the covalent cross-links in the vitrimer remain networked at both temperatures. On the other hand, $dc-(PE_{60}/PP_{40})-g-(D_5-co-S_2)$ can be readily degraded by diols (e.g., 1,2-propanediol), triggered by cleavage of the borate ester bond (Figure 1c3), by following the de-cross-linking pathway shown in Figure 1d.

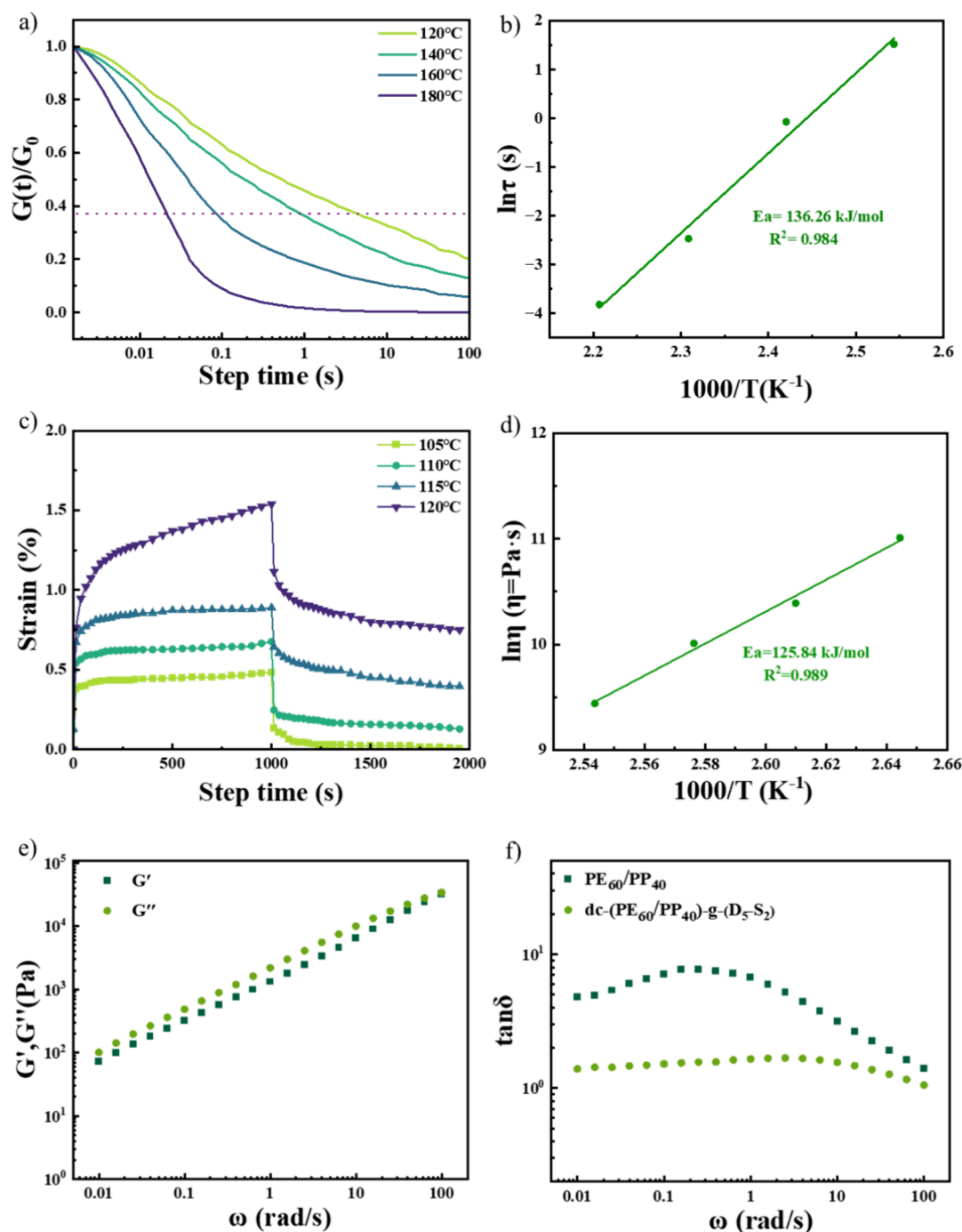


Figure 3. (a) Stress–relaxation curves of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) at different temperatures. (b) Arrhenius apparent activation energy of stress relaxation associated with average relaxation time t for dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂). (c) Creep and creep recovery curves of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) at different temperatures under a constant stress of 1 kPa. (d) Arrhenius activation energy of viscosity for dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) as calculated from creep data taken over a temperature range of 105–120 °C. (e) Relationship between storage modulus (G'), loss modulus (G''), and angular frequency (ω) of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) under dynamic frequency sweep at 180 °C. (f) Relationship between loss factor ($\tan \delta$) and angular frequency (ω) for PE₆₀/PP₄₀ and dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) under dynamic frequency sweep at 180 °C.

The effect of the composition of the PE₆₀/PP₄₀ blend on the mechanical properties of dc-(PE₆₀/PP₄₀)-g-(D_x-co-S_y) was investigated through tensile testing. After grafting with DBMA/St and cross-linking modification with 1,4-phenylenediboronic acid ester, the material exhibited increased fracture strength and elongation at break, alongside decreased Young's modulus (Figure 2a,b). dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂), with the loadings of DBMA, St, and 1,4-phenylenediboronic acid ester being 5, 2, and 0.5 wt %, respectively, exhibited

much improved tensile strength by 102.57% (from 19.03 to 38.55 MPa) and elongation at break surged by 1422.22% (from 63% to 896%), with 5.95% slightly decreased Young's modulus (from 328.42 to 308.87 MPa), compared to those of PP, achieving a balanced strengthening-toughening effect.

Further adjusting the content of DBMA or cross-linker has led to further tuned mechanical properties of dc-(PE₆₀/PP₄₀)-g-(D_x-co-S_y) (Table S1). As anticipated, with increasing DBMA or cross-linker contents, the chain cross-linking density

rose while molecular mobility became restricted due to the formation of a more confined network. Consequently, both Young's modulus and elongation at break decreased, leading to reduced fracture strength.

Commercial POE resin was employed to modify the PE₆₀/PP₄₀ blend at different loadings for improved mechanical properties (Figure S2 and Table S2). However, even optimal PE₆₀/PP₄₀/POE₁₅ is far inferior to its dynamically cross-linked structure-modified counterpart, dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) (Red vs green in Figure 2c). These results indicate that incorporating dynamic borate ester bonds offers a facile cross-linking strategy of tailoring the strength-toughness balance of the PE₆₀/PP₄₀ blend.

The thermal characterizations of polymeric materials determine their performance and processability. DSC was employed to investigate the crystallization behavior of PE₆₀/PP₄₀ blend and dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) (Figure 2d,e). 102.9 and 114.5 °C were identified as the crystallization temperatures of PE and PP (T_{cPE} and T_{cPP}) in PE₆₀/PP₄₀ blend, respectively, while both values shifted lower (100.7 and 113.4 °C) for dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂). The introduction of the dynamic cross-linking network restricts molecular chain mobility and suppresses the formation of highly ordered crystalline lattices in long ranges for both PE and PP, leading to lower crystallization temperatures. The respective melting points of PE and PP (T_{mPE} and T_{mPP}) in the PE₆₀/PP₄₀ blend were 117.4 and 160.5 °C, with crystallinities of 36.3% and 40.1%, respectively. All these values were lower in dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) (Table S3). Dynamic cross-linking-induced crystallinity reduction was further confirmed by wide-angle X-ray diffraction (WAXD) (Figure S3 and Table S4).

SEM images of cryo-fractured surfaces (Figure 2f) revealed distinct phase interfaces in PE₆₀/PP₄₀, where PP dispersed unevenly as coarse domains throughout the PE matrix—a typical characteristic of incompatible blends leading easily to performance degradation. In contrast, incorporation of boronic ester-based dynamic cross-links reduced the size of PP domains, improved dispersion uniformity, thereby compatibilizing the PE–PP interfaces. This structural optimization minimized interfacial voids and defects, enhanced interfacial adhesion, and ultimately improved phase compatibility, all were consistent with the mechanical property enhancements observed from tensile tests.

The nature of the dynamic covalent network of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) was investigated through stress relaxation experiments. Within the linear viscoelastic region, a constant strain of 1% was applied to the polymer at elevated temperatures (Figure 3a and Figure S4). Numerous studies on dynamic covalent networks assume that the materials obey a Maxwell model with single-mode exponential decay, thus exhibiting a single characteristic relaxation time (τ).^{30,31} Based on this assumption, τ was defined as the time required for the relaxation modulus to decay to 1/e ($\approx 36.8\%$) of its initial value. When the temperature was increased from 120 to 180 °C, τ for dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) decreased from 4.572 to 0.022 s, slightly longer than that of PE₆₀/PP₄₀ (0.471 to 0.018 s, Table S5). The relaxation time of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) closely resembles that of PE₆₀/PP₄₀ as expected, due to the plastic behavior of the network at elevated temperatures. Significant relaxation during processing can be anticipated for dynamic covalent networks. The temperature-dependent correlation of τ was found to reasonably obey the Arrhenius relationship. The stress relaxation activation energies ($E_{a,\tau}$) of

dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) and PE₆₀/PP₄₀ are 136.26 kJ·mol⁻¹ and 81.76 kJ·mol⁻¹, respectively (Figure 3b and Figure S5). The difference in $E_{a,\tau}$ arises from the fact that the relaxation of PE₆₀/PP₄₀ relies exclusively on molecular chain disentanglement (lower $E_{a,\tau}$), whereas the temperature-sensitive relaxation of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) is attributed to exchange reactions among dioxaborolane dynamic bonds (higher $E_{a,\tau}$).

The dissociation-reconstruction characteristics of the dynamic covalent network are also reflected in the creep and creep recovery experiments of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂). Under a constant stress of 1 kPa at 105 °C for 1000 s, the creep strain of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) (0.48%, Figure 3c) is significantly lower than that of PE₆₀/PP₄₀ (1.35%, Figure S6), indicating superior dimensional stability and creep resistance of the dynamically cross-linked material. This anticreep behavior is attributed to the frozen dioxaborolane dynamic network, which enhances local cross-linking density and restricts chain mobility. When the stress was removed at 105 °C, the strain recovered almost completely within 1000 s (0.0058%, Figure 3c). However, the residual strain of PE₆₀/PP₄₀ remained at 0.3989% (Figure S6). These results demonstrate the extremely slow relaxation of the chemical structure of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) at 105 °C, minimal creep, and rapid recovery, all outperforming PE₆₀/PP₄₀ in creep resistance.

As temperature increases, the anticreep performance of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) weakens due to accelerated topological rearrangement via dynamic boronate bond exchanges. Higher temperatures promote cross-link reshuffling and chain migration, thereby balancing creep resistance and processability. According to the Arrhenius plot of temperature-dependent viscosity derived from the creep data,^{31,32} the activation energies for viscous flow ($E_{a,\eta}$) of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) and PE₆₀/PP₄₀ are 125.84 kJ·mol⁻¹ and 71.92 kJ·mol⁻¹, respectively (Figure 3d and Figure S7). These align with stress relaxation results, further confirming the high-temperature sensitivity and dynamic dissociation-reconstruction nature of the network in dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂).

Dynamic frequency sweep tests were conducted on PE₆₀/PP₄₀ and dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂). It can be observed that the loss modulus (G'') and storage modulus (G') of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) are nearly parallel and closely matched within the tested angular frequency range (0.01–100 rad/s) (Figure 3e), a pattern distinctly different from that of PE₆₀/PP₄₀ (Figure S8). The loss factor of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) only decreases at high frequencies (Figure 3f), indicating an almost constant viscoelastic ratio. The presence of the dynamically cross-linked network significantly restricts molecular chain motion and relaxation. Even at low frequencies, the chains cannot flow freely and disentangle like a linear melt. The entire material behaves more like a weakened elastic network or gel, as dominated primarily by the elasticity, with viscous flow remarkably suppressed.

The mechanical properties of thermoplastic polymers deteriorate with increasing reprocessing cycles, due primarily to thermomechanical degradation induced by repeated processing. Dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂), on the other hand, is anticipated to exhibit superior recyclability through boroxane bond exchange-induced network rearrangement, i.e., dynamic boronic ester bond enabled reversible transesterification. After grinded into fragments/powders, dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂) can readily be remolded into dumbbell-shaped specimens

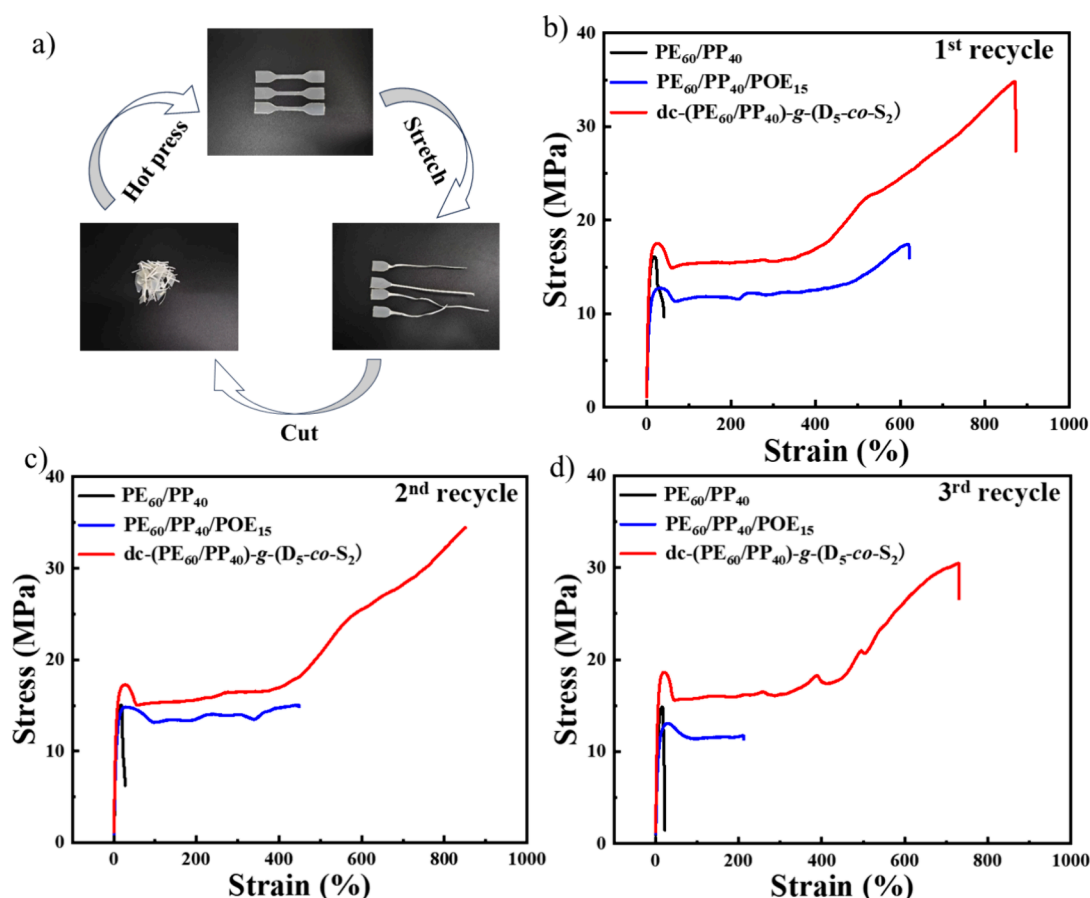


Figure 4. (a) Thermo-mechanical cyclic processing. Stress–strain curves of $\text{PE}_{60}/\text{PP}_{40}$, $\text{PE}_{60}/\text{PP}_{40}/\text{POE}_{15}$, and $\text{dc}-(\text{PE}_{60}/\text{PP}_{40})\text{-g}-(\text{D}_5\text{-co-S}_2)$ after the (b) 1st, (c) 2nd, and (d) 3rd cycle of processing.

(Figure 4a). After multiple reprocessing cycles, the Young's modulus, tensile strength, and elongation at break of $\text{dc}-(\text{PE}_{60}/\text{PP}_{40})\text{-g}-(\text{D}_5\text{-co-S}_2)$ consistently surpass those of $\text{PE}_{60}/\text{PP}_{40}$ and $\text{PE}_{60}/\text{PP}_{40}/\text{POE}_{15}$ for each recycle, Figure 4b–d, maintaining high-performance regardless the number of cycles.

After three reprocessing cycles, the mechanical properties of both $\text{PE}_{60}/\text{PP}_{40}$ and $\text{PE}_{60}/\text{PP}_{40}/\text{POE}_{15}$ have significantly deteriorated (Table S6). The tensile strength of $\text{PE}_{60}/\text{PP}_{40}$, for example, dropped to 14.86 MPa from the original 19.03 MPa, with a reduction in elongation at break by 65% (63.4% to 22.1%). For $\text{PE}_{60}/\text{PP}_{40}/\text{POE}_{15}$, the tensile strength and elongation at break declined to 13.12 MPa and 199% from the original 22.65 MPa and 717%, respectively. In contrast, $\text{dc}-(\text{PE}_{60}/\text{PP}_{40})\text{-g}-(\text{D}_5\text{-co-S}_2)$ maintained high tensile strength of 31.49 MPa and elongation at break of 730% after three cycles. Besides, the Young's modulus of $\text{dc}-(\text{PE}_{60}/\text{PP}_{40})\text{-g}-(\text{D}_5\text{-co-S}_2)$ was as high as 291.08 MPa (94.29% retention), whereas $\text{PE}_{60}/\text{PP}_{40}$ and $\text{PE}_{60}/\text{PP}_{40}/\text{POE}_{15}$ were 266.05 MPa (81.01% retention) and 173.54 MPa (84.51% retention), respectively. The serious performance deteriorations of $\text{PE}_{60}/\text{PP}_{40}$ and $\text{PE}_{60}/\text{PP}_{40}/\text{POE}_{15}$ during thermal recycling stem primarily from varying degrees of degradation, including chain scission, branching, and de-cross-linking, and the inherent PE–PP incompatibility in the two materials. Unlike either, $\text{dc}-(\text{PE}_{60}/\text{PP}_{40})\text{-g}-(\text{D}_5\text{-co-S}_2)$ consistently maintains a dynamic dissociation–exchange state during thermal cycling, benefiting from the dynamic covalent network. The dissociation and reconstruction of boroxane bonds help keep the structural integrity of the dynamic network by enabling its rearrangement and preventing

backbone cleavage from taking place. Entropically, the boroxane bonds that we deliberately introduced into $\text{PE}_{60}/\text{PP}_{40}$ act as heat sinks or thermo-chemical energy converters, effectively bypassing the damage that would otherwise have been caused to PE or PP polymers *per se*, the key to the success of our strategy.

3. CONCLUSIONS

In summary, we report a novel approach to effective compatibilization polyolefin blends by constructing a dynamic boron–oxygen covalent network within $\text{PE}_{60}/\text{PP}_{40}$, a representative model PE and PP blend, developing a thermochemically robust, yet recyclable $\text{dc}-(\text{PE}_{60}/\text{PP}_{40})\text{-g}-(\text{D}_5\text{-co-S}_2)$ hybrid plastic. $\text{Dc}-(\text{PE}_{60}/\text{PP}_{40})\text{-g}-(\text{D}_5\text{-co-S}_2)$ exhibits significantly enhanced PE–PP interfacial adhesion with much smaller PP droplet sizes and narrower size distributions. Additionally, incorporation of the dynamic boron–oxygen covalent network endows $\text{dc}-(\text{PE}_{60}/\text{PP}_{40})\text{-g}-(\text{D}_5\text{-co-S}_2)$ with remarkably improved mechanical properties. This approach may have provided a highly feasible and scalable strategy of compatibilizing polyolefin blends in industry.

4. EXPERIMENTAL SECTION

4.1. Materials. Exxon Mobil Chemical, United States, provides the linear low density polyethylene (LLDPE) resin Exceed 3518PA, which is an ethylene 1-hexene copolymer with a melt flow rate (MFR) of $3.5 \text{ g} \cdot 10 \text{ min}^{-1}$ (190 °C, 2.16 kg) and a density of $\rho = 920 \text{ kg} \cdot \text{m}^{-3}$. Taiwan Chemical Fiber Co., Ltd. provided isotactic polypropylene (iPP) resin K1P38 with a melt flow rate (MFR) of $38 \text{ g} \cdot 10 \text{ min}^{-1}$ (230

°C, 2.16 kg) and a density of $\rho = 920 \text{ kg}\cdot\text{m}^{-3}$. Exxon Mobil Chemical Company, United States, provides the polyolefin elastomer (POE) resin Vistamaxx 6102FL, with a melt flow rate (MFR) of $1.4 \text{ g}\cdot 10 \text{ min}^{-1}$ (190 °C, 2.16 kg) and a density of $\rho = 862 \text{ kg}\cdot\text{m}^{-3}$. 2,3-Dihydroxypropyl methacrylate (GMMA), phenylboronic acid, 1,4-phenylenediboronic acid, and styrene were obtained from Shanghai Macklin Biochemical Co., Ltd. 1,2-Propanediol and dicumyl peroxide (DCP) were obtained from Sinopharm Chemical Reagent Co., Ltd. Functional monomer dioxaborolane monomethacrylate (DBMA) and arylboronic ester cross-linker 1,4-phenylenediboronic acid were prepared by the Polymer Engineering Laboratory of Hubei University of Technology according to the literature.^{33,34}

4.2. Preparation of PE/PP Hybrid Plastic. Melt grafting experiments were carried out in a Rheocord Haake batch mixer (RM-200C, Hap Electric Technology Co., Ltd.). Unless otherwise specified, all samples in this experiment were prepared using a screw rotating speed of $30 \text{ r}\cdot\text{min}^{-1}$ and a batch mixer temperature of 170 °C. In brief, 30 g PE and 20 g PP were dry-blended with DBMA, styrene, and DCP before being introduced to the chamber of the batch mixer, with DCP and antioxidant both at 0.2 wt %. After 10 min, a sample of the blend was extracted from the chamber and quenched in cold water, terminating the reaction and resulting in a DCP and styrene-grafted PE₆₀/PP₄₀ blend, denoted as (PE₆₀/PP₄₀)-g-(D_x-co-S_y), where the subscripts x and y represent the weight percentages of DBMA and styrene, respectively. (PE₆₀/PP₄₀)-g-(D_x-co-S_y) was then dried at 85 °C for 12 h, prior to being mixed with 1,4-phenyldiboronic acid ester and the mixture being introduced to the batch mixer for 10 min (to complete dynamic cross-linking), resulting in a PE₆₀/PP₄₀-based vitrimer, denoted as dc-(PE₆₀/PP₄₀)-g-(D_x-co-S_y). Dc-(PE₆₀/PP₄₀)-g-(D_x-co-S_y) was then shaped into thin plates of 100 mm long $\times$ 100 mm wide $\times$ $1 \pm 0.1 \text{ mm}$ thick using a plate vulcanization machine (YJ-46, Chengdu Hangfa Hydraulic Engineering Co., Ltd., China) at 180 °C and 10 MPa for further characterizations, measurements, and tests. A blend of PE and PP of the same PE/PP weight ratio was prepared in the same way, only without any other components, denoted as PE₆₀/PP₄₀.

4.3. Measurements and Characterizations. Tensile tests were conducted according to the GB/T10402–2006 standard using an electronic tensile tester (GMT4000, GTS systems Co., Ltd., China), at a temperature of 25 °C, and with an extension rate of $100.0 \text{ mm}\cdot\text{min}^{-1}$. Dumbbell-shaped specimens (width of 4 mm, length of 75 mm, and thickness of 1 mm) were prepared using a mold and an electric punching machine (CP-25-III, Yangzhou Daochun Testing Machinery Co., Ltd., China). Elongation at break (%), tensile strength (MPa), and Young's modulus (MPa) of each specimen were measured with at least five samples. Elongation at break refers to the tensile strain of the specimen at fracture. Tensile strength refers to the maximum tensile force per unit area that causes the sample to break. Young's modulus is calculated from the slope within the initial linear region of the stress–strain curve.

Fourier transform infrared spectroscopy (FTIR) analysis was conducted using the potassium bromide (KBr) pellet method. The sample was mixed with spectroscopic-grade KBr powder, ground uniformly, and pressed into transparent pellets. Infrared spectra were scanned in the range of $4000\text{--}500 \text{ cm}^{-1}$ to obtain absorption characteristics.

The crystallization behavior of the samples was studied with a differential scanning calorimeter (DSC8000, PerkinElmer, United States) in N₂ atmosphere with a nitrogen flow rate of $20 \text{ mL}\cdot\text{min}^{-1}$. The original sample tests were divided into heating, cooling, and reheating stages, the first temperature ramp was to eliminate thermal histories of the samples. The heating and cooling rates of all samples were set to $10 \text{ }^\circ\text{C}\cdot\text{min}^{-1}$ and the scanning range was $20\text{--}200 \text{ }^\circ\text{C}$.

For scanning electron microscope (SEM, SU8010, Hitachi, Japan) imaging, all samples were injection-molded into notched specimens of 5 mm in width and thickness, then immersed in liquid nitrogen for 2 h to undergo brittle fracture. The fractured surfaces were sputter-coated with gold to obtain SEM-compatible surfaces prior to the examination of the micromorphology of the fracture surfaces.

Small-amplitude oscillatory shear (SAOS), shear creep, creep-recovery, and stress relaxation experiments were carried out on a rotational rheometer (DHR-2, TA Instruments, USA) equipped with a parallel-plate (25 mm in diameter with a gap of 1.0 mm) under nitrogen atmosphere. SAOS was also used to perform rheological tests for melts. The change in rheological properties with frequency was observed with a dynamic frequency sweep test performed in the range of $0.01\text{--}100 \text{ rad}\cdot\text{s}^{-1}$ with a predetermined strain amplitude of 1%.

Shear stress relaxation was characterized using the transient mode of a rotational rheometer at $120\text{--}180 \text{ }^\circ\text{C}$. Samples were allowed to be equilibrated at the test temperature for 10 min prior to the test. Once thermal equilibrium was reached, the samples were subjected to an instantaneous 1% strain, which was maintained throughout the test. The stress relaxation modulus was recorded until it completely relaxed.³¹

Shear creep and creep-recovery experiments were characterized using the transient mode of a rotational rheometer from 105 to 120 °C. The samples were equilibrated at the test temperature for at least 10 min prior to the experiment. For creep-recovery tests, a 1 kPa stress was applied for a period of 1000 s, followed by 1000 s of recovery (zero stress). The network viscosity (η) was calculated as a function of temperature as follows:³¹

$$\eta = \frac{\sigma}{\dot{\gamma}} \quad (1)$$

where σ is the creep shear stress and is the shear strain rate calculated from the fitted slope of the linear part of the creep curves.

The mechanical tests of recycled samples were carried out in essentially the same way as the tensile tests described above. In brief, the fractured samples were cut into pieces, then processed in a torque rheometer under processing conditions of 170 °C and 30 rpm for 10 min. Next, the samples were molded into a plate with dimensions of $100 \text{ mm} \times 100 \text{ mm}$ and a thickness of $1 \pm 0.1 \text{ mm}$ using a plate vulcanizer under conditions of 180 °C and 10 MPa. Finally, dumbbell-shaped specimens with a width of 4 mm and a thickness of 1 mm were prepared using an electric punch cutter for tensile testing. The above operations were repeated three times to obtain satisfactory statistics.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.macromol.5c02898>.

Gel content of dc-(PE₆₀/PP₄₀)-g-(D₅-co-S₂); compatibilization of PE₆₀/PP₄₀/POE at different POE loadings; XRD spectra; rheological property of PE₆₀/PP₄₀; mechanical properties, DSC, and stress relaxation data sheet (PDF)

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Notes

The authors declare no competing financial interest.

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